Marketing Planning Assistant Agent Report

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement & Objectives

1.1 Problem Statement

Traditional marketing planning is a time-consuming, manual process that requires a team of specialists — a market researcher, a campaign strategist, a project scheduler, and a report writer — collaborating over several days to produce a single marketing campaign document. Small businesses, startups, and individual marketers often lack the resources to maintain such a team, resulting in either poorly planned campaigns or delayed execution. There was a clear need for an intelligent, automated system that could replicate this entire workflow and deliver a structured, professional marketing plan quickly and affordably.

Key problem areas:

- To automate the end-to-end process of marketing campaign planning using Agentic AI.
- To reduce the time required to generate a complete marketing plan from several days to under 5 minutes.
- To build a multi-agent system where each agent has a specific, specialized role — planning, research, strategy, scheduling, and reporting.
- To produce a structured, 8-section executive marketing plan report as the final output.
- To provide a simple, accessible web interface where any user can input a marketing goal and receive a full plan without requiring technical expertise.

1.2 Project Objectives

The core objectives of the Marketing Planning Assistant Agent are:

Automation — To automate the complete end-to-end marketing campaign planning process using five specialized AI agents.

Speed — To reduce plan generation time from several days to under 5 minutes using Groq's high-speed LLM inference.

Specialization — To assign distinct roles to each agent — Planner, Researcher, Strategist, Scheduler, and Reporter — for focused, expert-level output at every stage.

Collaboration — To build a context-passing pipeline where each agent's output feeds directly into the next, ensuring a coherent and connected final plan.

Accessibility — To make professional marketing planning available to non-technical users through a simple Streamlit web interface.

Affordability — To build a fully functional Agentic AI system using only free-tier tools — CrewAI, Groq API, and Streamlit.

1.3 Scope of the Project

The scope of this project covers the design and development of a fully functional Agentic AI system for marketing campaign planning. The system accepts a natural language marketing goal from the user along with campaign parameters such as budget, duration, and target market. It then runs five AI agents sequentially — Planner, Research, Strategy, Scheduler, and Reporter — each building on the output of the previous one. The final deliverable is a downloadable marketing plan saved as a text file. The project is developed as a prototype using free-tier tools and is scoped for single-user, local execution. Database integration, multi-user support, and production deployment are outside the current scope but identified as future enhancements.

- The system is designed to handle any marketing goal expressed in natural language, making it domain-agnostic — it can plan campaigns for product launches, service promotions, brand awareness, or local business marketing.

- The scope includes the design of 11 specialized mock tools that simulate real-world data sources for competitor research, audience insights, budget calculation, timeline generation, and risk assessment.

- The project covers the complete user journey — from entering a goal on the web interface, through real-time agent execution with live status updates, to downloading the final marketing plan as a text file.

- The system is designed to be stateless and session-based, meaning no user data is stored between runs, keeping the prototype lightweight, private, and easy to manage.

- The scope explicitly excludes real-time internet search, live competitor data APIs, multi-user authentication, and cloud deployment — these are identified as future enhancements beyond the current prototype phase.

2. Proposed Solution

2.1 Key Features

The Marketing Planning Assistant Agent is a multi-agent AI system that automates the entire marketing campaign planning process from a single user-defined goal. It replaces the manual effort of a full marketing team by running five specialized AI agents sequentially, each contributing a distinct layer of intelligence to the final output.

Key features of the system include:

Multi-Agent Collaboration — Five specialized agents (Planner, Research, Strategy, Scheduler, Reporter) work sequentially, each passing full context to the next.

Automated Market Research — Competitor profiling, audience persona building, and market trend identification done automatically.

Campaign Strategy Generation — AI-generated positioning statement, channel mix, budget allocation, and 7 measurable KPIs with targets and deadlines.

Execution Timeline — A 12-week phased plan divided into Pre-Launch, Launch, Optimize, and Evaluate stages with task owners and dependencies.

Executive Report — A complete 8-section marketing plan report including risks, quick wins, and appendix, generated and saved as a downloadable file.

Real-Time UI Feedback — Live agent status tracking on Streamlit with colour-coded progress indicators (grey → yellow → green).

2.2 Overall Architecture / Workflow

The system follows a layered, modular multi-agent architecture. At the top is the Frontend layer — a Streamlit web application where the user inputs their marketing goal and campaign settings. This connects to the Backend layer, which is the MarketingCrew module (crew.py) responsible for orchestrating the entire pipeline.

Here is the Architecture Diagram:

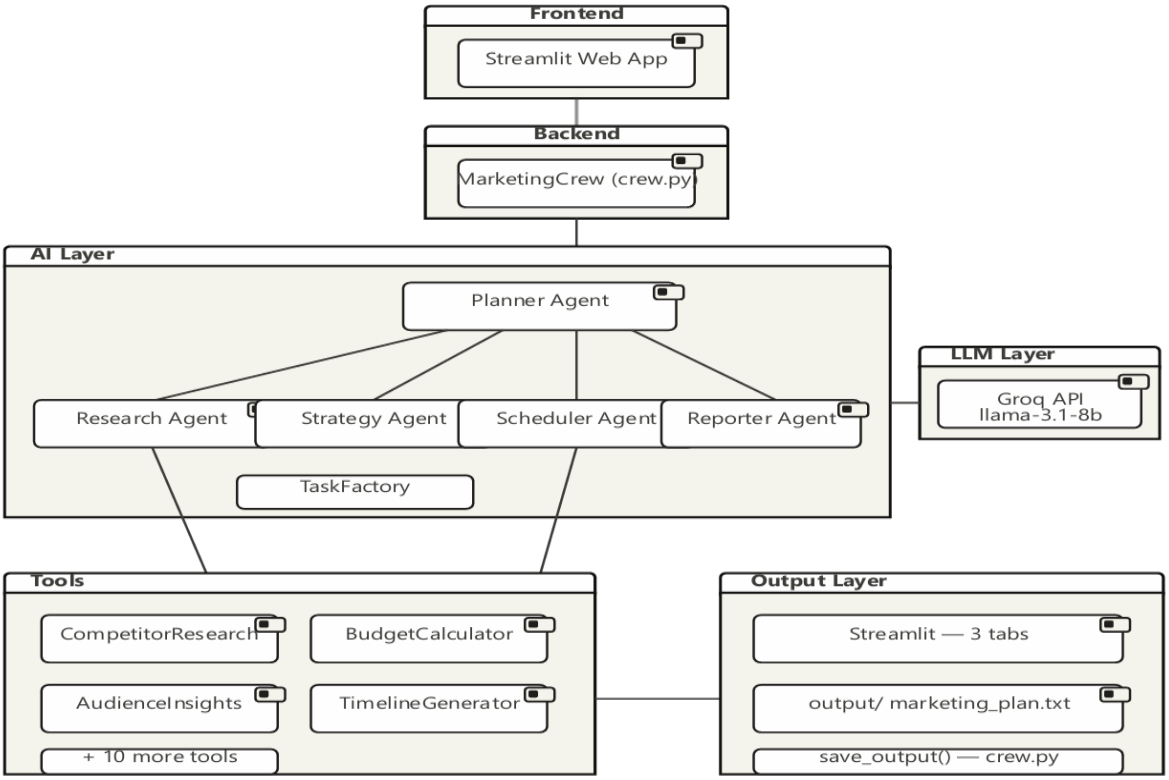

Figure 1 Architecture Diagram

Workflow

1. Frontend Layer — Streamlit Web App The user enters their marketing goal along with campaign parameters such as budget, duration, and target market. This layer captures input, displays real-time agent progress, and presents the final output across 3 tabs.

2. Backend Layer — MarketingCrew (crew.py) The submitted goal is passed to the MarketingCrew module — the central orchestration unit that initializes the pipeline,builds all five agents and their tasks, kicks off sequential execution, and saves the final output.

3. AI Layer — Agents & TaskFactory This is the core intelligence layer of the system. The Planner Agent acts as the entry point — it receives the goal, decomposes it into 8 prioritized sub-tasks, and delegates them to the four specialized agents:

- **Research Agent** — analyzes competitors, builds audience personas, and identifies market trends
- **Strategy Agent** — defines the channel mix, budget allocation, and 7 measurable KPIs

- **Scheduler Agent** — converts the strategy into a 12-week phased execution timeline
- **Reporter Agent** — aggregates all previous outputs and writes the complete 8-section executive report

4. LLM Layer — Groq API / llama-3.1-8b All reasoning and content generation across every agent is handled by the Groq API running the llama-3.1-8b model at temperature 0.3 with a max token limit of 600 per agent call.

5. Tools Layer 11 mock tools including CompetitorResearch, AudienceInsights, BudgetCalculator, and TimelineGenerator support agents during execution to validate data, calculate budgets, and assess risks.

6. Output Layer Results are displayed on Streamlit across 3 tabs and saved as a timestamped marketing_plan.txt file via the save_output() function in crew.py.

The overall architecture flow is:

Figure 2 Workflow Diagram

2.3 Tools & Technologies Used

Layers	Technology	Purpose
Programming Language	Python 3.12	Core development language
Multi-Agent Framework	Crew AI	Agent orchestration and sequential task execution
LLM Provider	Groq API	Free-tier LLM inference
LLM Model	llama3-8b-8192 (LLaMA 3.1 8B Instant)	Language generation for all agents
Web Interface	Streamlit	Frontend UI with real-time agent status updates
Supporting Library	LangChain	LLM tooling and prompt management
API Security	python-dotenv (.env file)	Secure API key storage
Output Format	Plain text (.txt)	Timestamped downloadable marketing plan
Mock Tools	Custom Python classes (BaseTool ×11)	Competitor research, budget calculation, KPI generation, timeline generation, risk assessment, etc.

3. Results & Output

Screenshots / Outputs

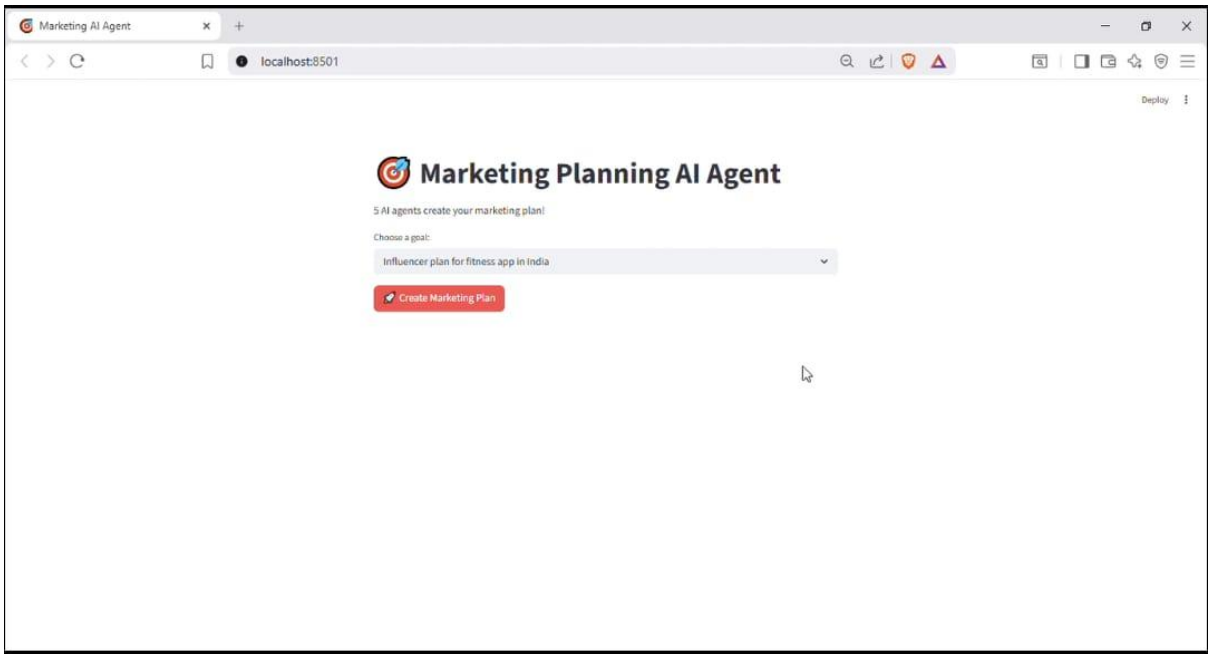

Figure 3: User Interface of Dashboard

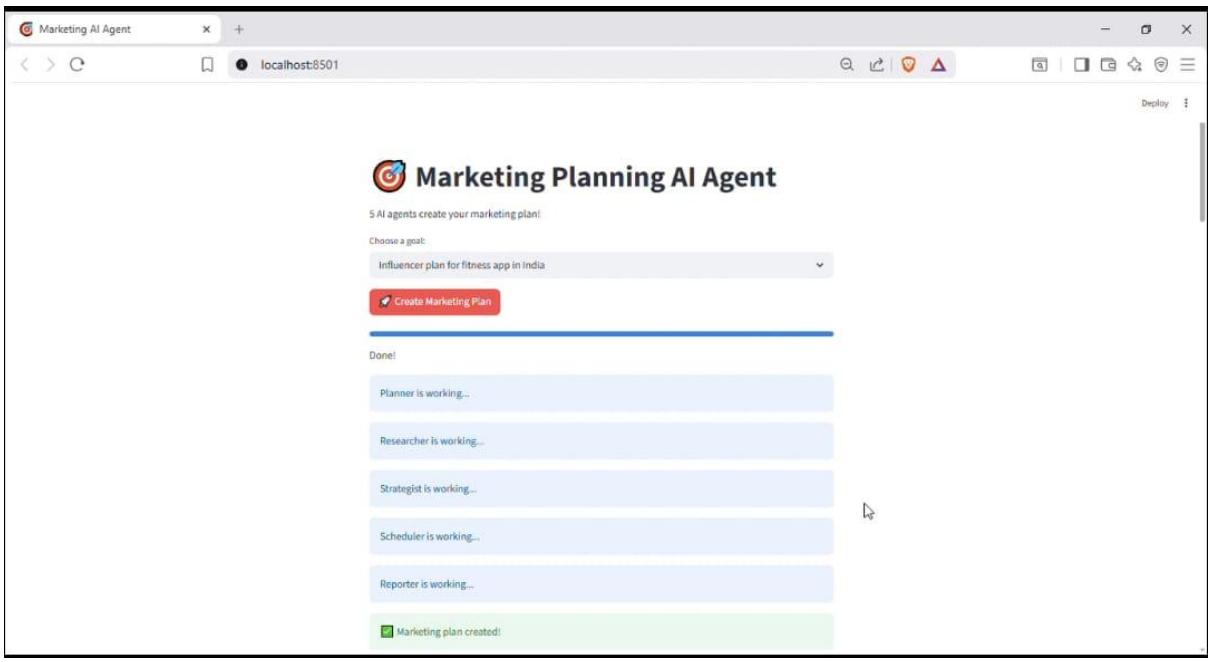

Figure 4: User Input with Five Agents Processing

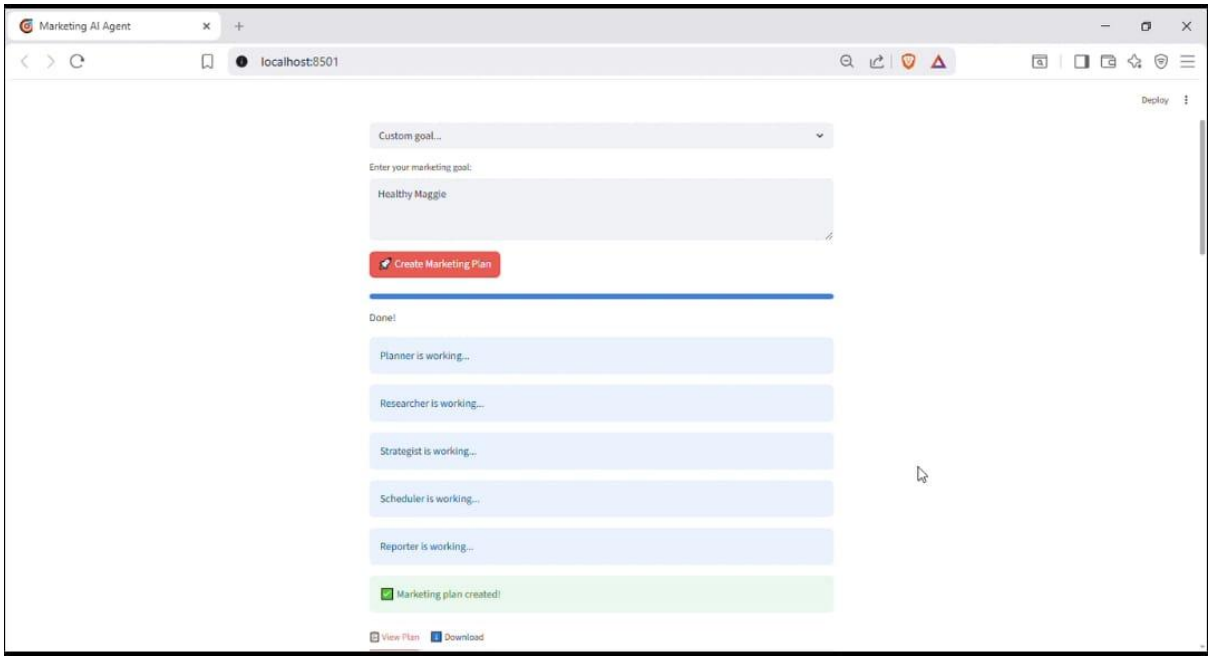

Figure 5: Manual Input with Five Agents Processing

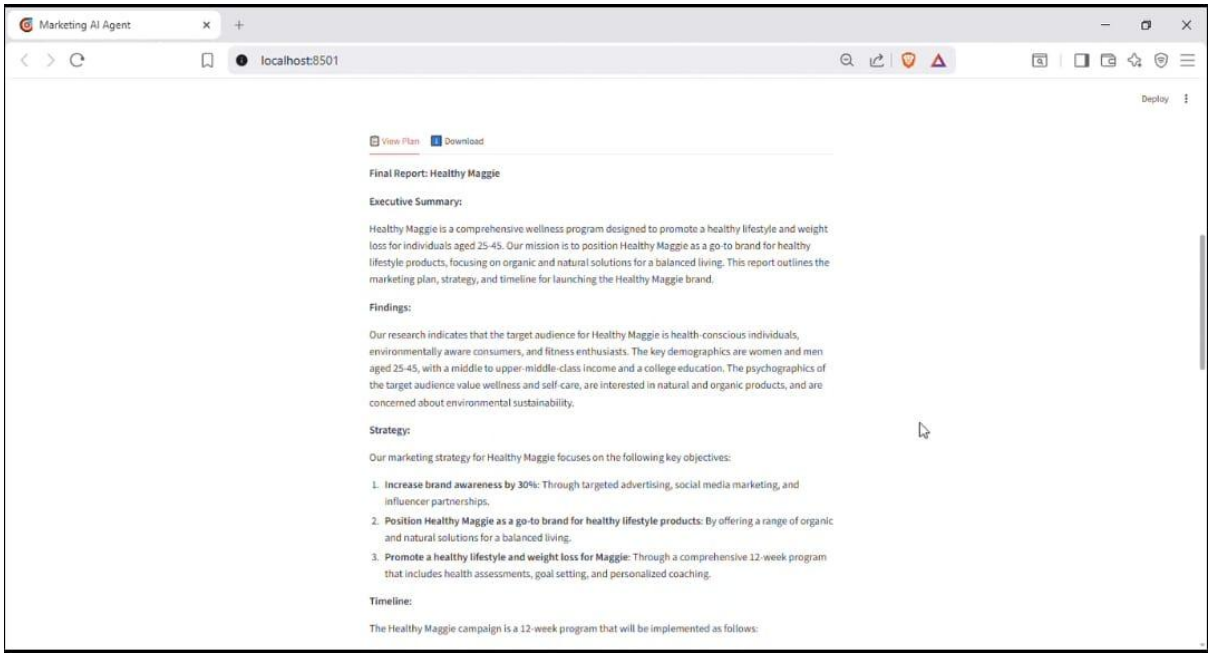

Figure 6: Generated Output Interface

Key Outcomes

The Marketing Planning Assistant Agent successfully proved that a five-agent AI pipeline can replace a full marketing team's planning effort and deliver a structured, professional campaign plan in under 5 minutes. This validated that free-tier tools like CrewAI, Groq, and Streamlit are fully capable of building real-world Agentic AI solutions.

- A complete, structured marketing campaign plan was generated from a single natural language goal in under 5 minutes.
- The system successfully demonstrated sequential multi-agent collaboration, with each agent's output serving as context for the next.
- The 5-agent pipeline produced competitor analysis, 2 audience personas, 5 market trends, a full channel strategy, budget allocation, 7 KPIs, a 12-week phased timeline, and a risk assessment — all automatically.
- The Streamlit interface provided real-time visibility into agent execution with live status updates and expandable outputs.
- The final marketing plan was saved as a downloadable .txt file, making it immediately usable outside the application.

4. Conclusion

The Marketing Planning Assistant Agent successfully demonstrates the practical application of Agentic AI in automating a complex, multi-step real-world workflow. By leveraging CrewAI's multi-agent orchestration framework, Groq's free-tier LLM inference, and Streamlit's interactive UI, the project replaced a traditionally manual, multi-day marketing planning process with an intelligent pipeline that delivers a professional campaign plan in under five minutes.

The key learning from this project was understanding how to design agents with clear, specialized roles and how to pass contextual information between them sequentially to build progressively richer outputs. We also learned how to integrate LLM APIs within a structured agentic framework, design mock tools to simulate real-world data sources, and build a user-facing interface that makes AI outputs accessible and actionable.

5. Future Scope & Enhancements

Database Integration: Connecting PostgreSQL to store run history and ChromaDB for semantic search across past marketing plans.

Parallel Agent Execution: Running agents concurrently using Celery + Redis to reduce total runtime from 4–6 minutes to approximately 2 minutes.

Multi-user Support: Adding authentication (JWT/OAuth2) and user-specific session management to support team-level usage.

Production Deployment: Migrating from Streamlit to a FastAPI + React architecture for a scalable, production-grade web application with HTTPS and API rate limiting.

Real API Integration: Replacing mock tools with live data sources — actual competitor intelligence APIs, real-time market trend feeds, and live budget benchmarking tools.

Export Formats: Supporting PDF and DOCX export in addition to the current .txt output.

Expanded Agent Capabilities: Adding a Content Creation Agent to auto-generate ad copy, social media posts, and email sequences as part of the campaign plan.